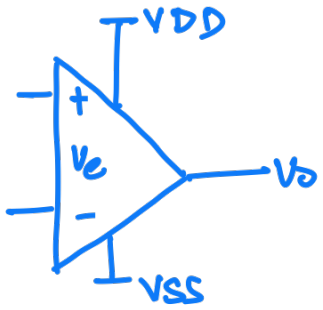
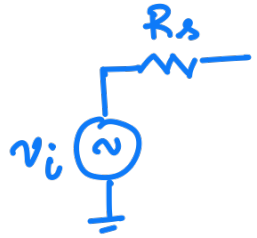
Lecture #04

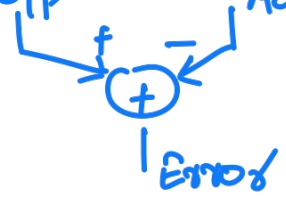


$V_o = A V_e$

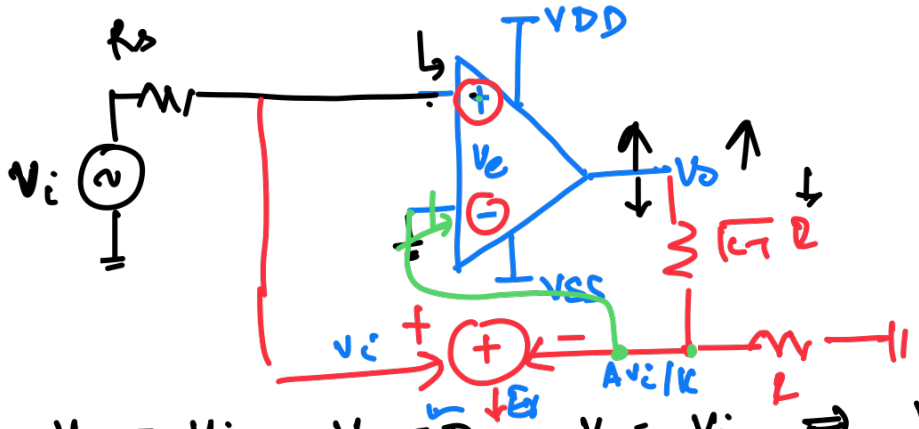


$\rightarrow V_o = K V_i$

Desired o/p (KV_i) — Actual o/p (KV_i)



$(KV_i) \downarrow V_i'$

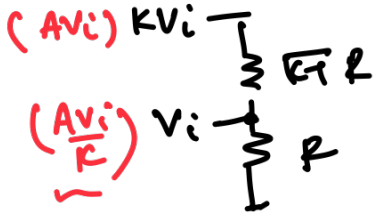


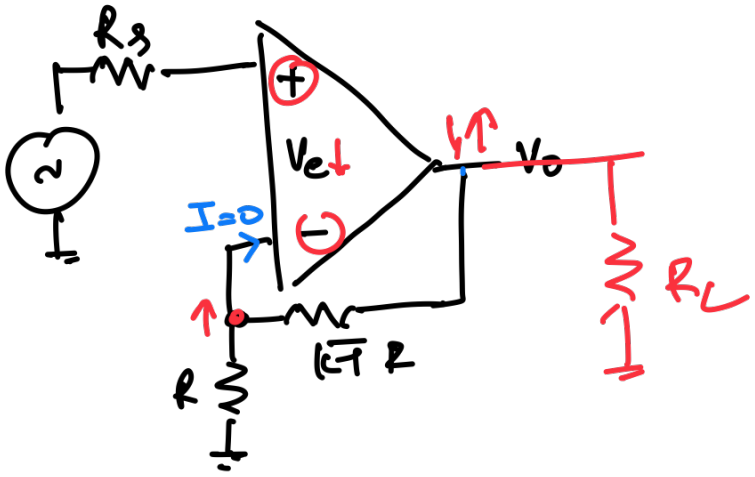
$V_+ = V_i \quad , V_- = 0 \quad , V_e = V_i \Rightarrow V_o = A V_i$

$V_- = \frac{A V_i}{K} \Rightarrow V_e = V_i - \frac{A V_i}{K} < 0$

$\downarrow$

$-A \left(\frac{A V_i}{K} - V_i \right)$





$V_o = A V_e$

$V_+ = V_i$, $V_- = \frac{V_o}{K}$

$V_e = \left(V_i - \frac{V_o}{K}\right)$

$V_o = A\left(V_i - \frac{V_o}{K}\right)$

$V_o\left(\frac{1}{A} + \frac{1}{K}\right) = V_i$

$V_- = \frac{KV_i/K}{1+\frac{K}{A}} = \frac{V_i}{1+\frac{K}{A}} \approx V_i$, $A \to \infty$

$V_o = \frac{KV_i}{1+\frac{K}{A}} \approx KV_i$, $A \gg K$

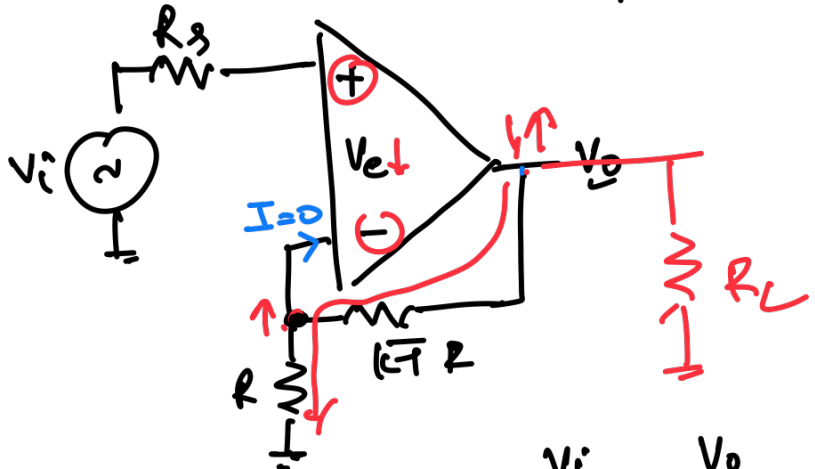
$K = 10$, $A = 10000$

if $A \to \infty$, $V_o = KV_i$

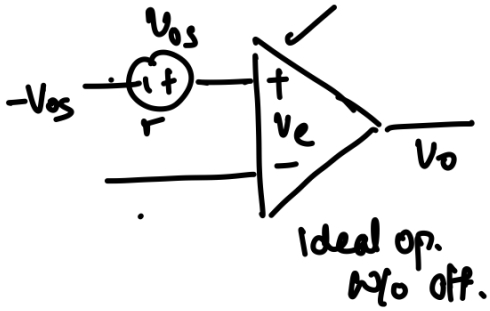
Ideal opamp.

- There is virtual short b/w V_+ and V_- (Ideal opamp)

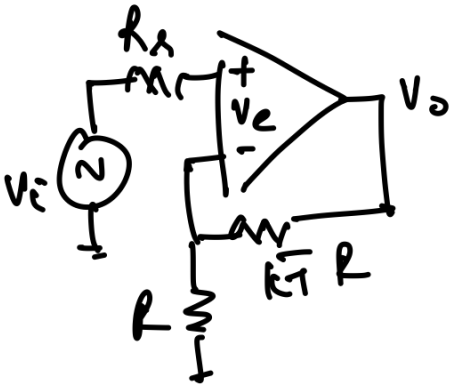
" Non-inverting amp. "



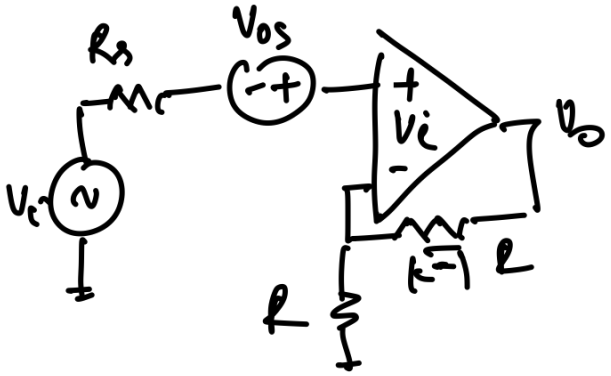
$V_+ = V_i$, $V_- = V_i$, $\frac{V_i}{R} = \frac{V_o}{KR}$ $\Rightarrow V_o = KV_i$



Ideal op. amp off.



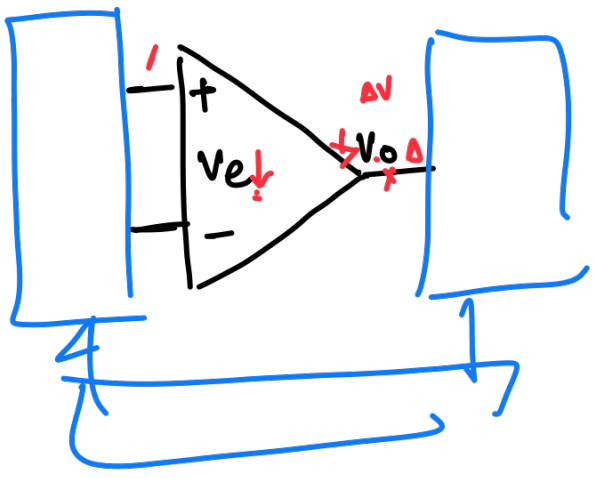
≡

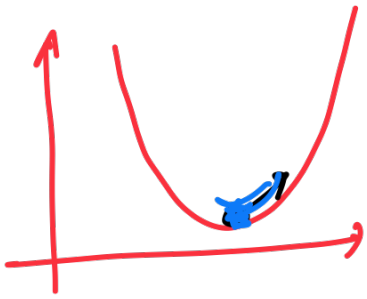


$$V_+ = V_i + V_{os} = V_-$$

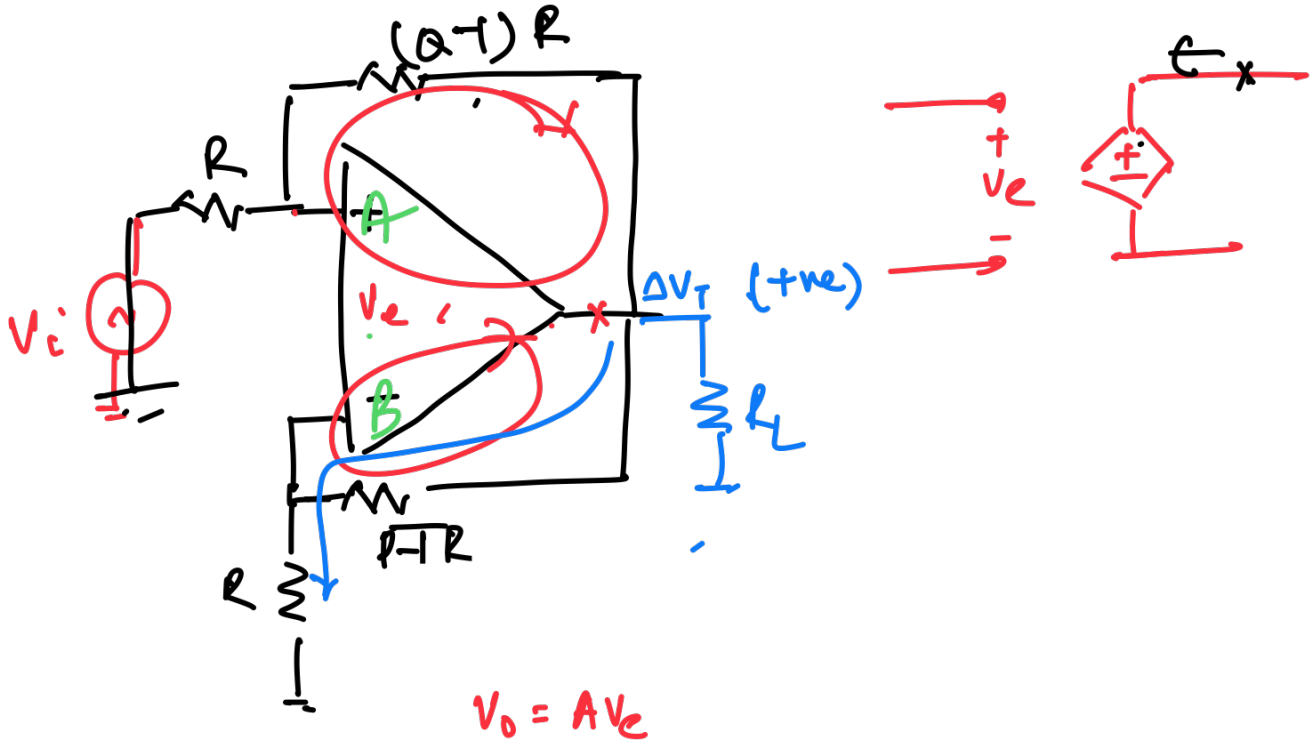
$$\frac{V_o}{KR} = \frac{V_i + V_{os}}{R}$$

$$V_o = KV_i + KV_{os}.$$





$$V_o = AV_e$$



$V_0 = A V_e$

$$V_- = \frac{1}{P}\Delta V_T$$

$$V_+ = \frac{1}{Q}\Delta V_T$$

$$V_e = V_T - V_- = \left(\frac{1}{Q} - \frac{1}{P}\right)\Delta V_T$$

$$= \frac{1-Q}{PQ}\Delta V_T$$

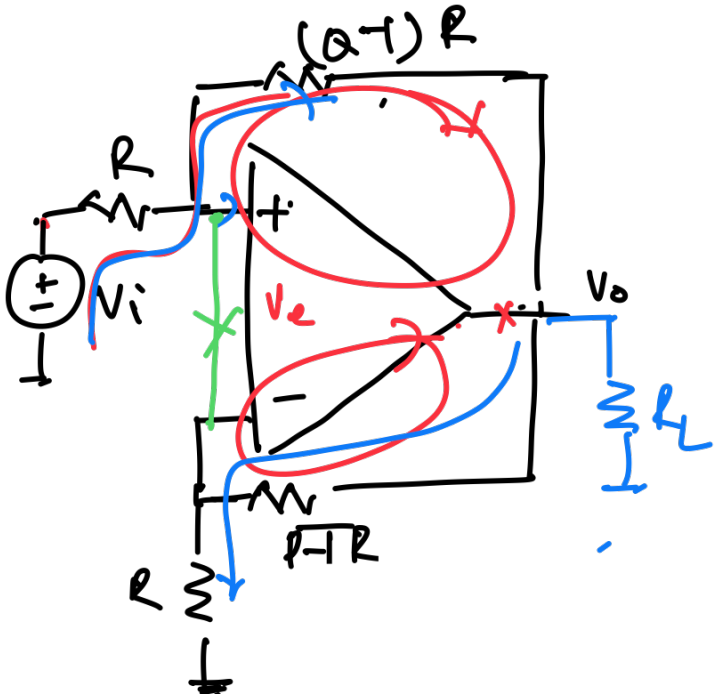
- Negative feedback if $\frac{V_e}{\Delta V_T} < 0 \Rightarrow V_+ = V_-$ when $A \to \infty$
- Positive feedback if $\frac{V_e}{\Delta V_T} > 0 \quad V_+ \neq V_-$

for negative feedback.

$$V_+ = V_- = \frac{V_0}{P}$$

$$\frac{V_i - V_-}{R} = \frac{V_- - V_0}{(Q-1)R}$$

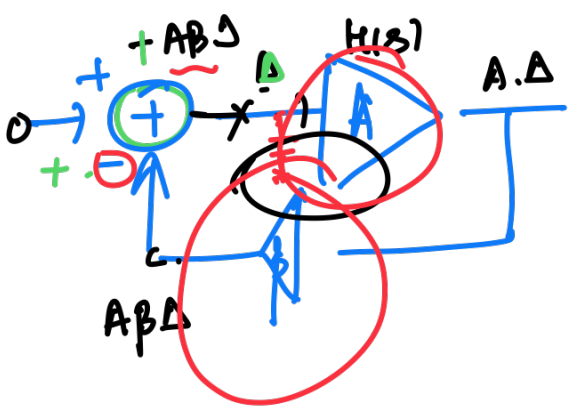
$$\left(V_i - \frac{V_0}{P}\right) = \frac{V_0}{P(Q-1)} - \frac{V_0}{Q-1}$$

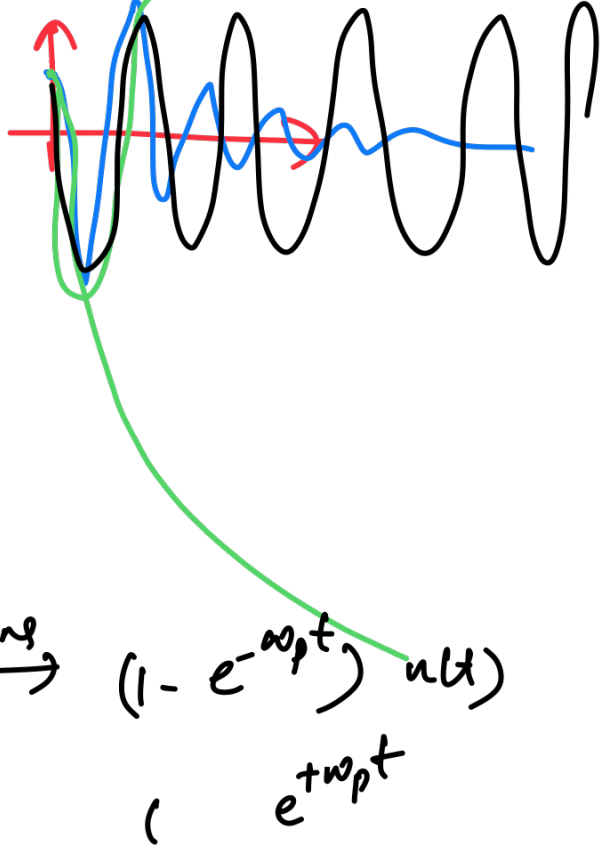


Final - Value Theo -

$$\lim_{t \to \infty} y(t) = \lim_{s \to 0} sY(s)$$

$Y(s) = H(s) \cdot U(s)$







$$H(s) = \frac{1}{1 + s/\omega_p} = \frac{1}{1 - s/\omega_p}$$

Step Response →

$$(1 - e^{-\omega_p t})\, u(t)$$

$$e^{+\omega_p t}$$